



VNU-HCM, UNIVERSITY OF SCIENCE
FACULTY OF PHYSICS AND ENGINEERING PHYSICS
DEPARTMENT OF THEORETICAL PHYSICS

UNDERGRADUATE THESIS

Variational Quantum Eigensolver

Speaker

NGUYỄN VŨ LINH



Contents

- 1. Introduction to quantum computing**
- 2. Transverse Ising model**
- 3. Variational quantum eigensolver (VQE)**
- 4. Implementation and results**
- 5. Quantum Chemistry with VQE**

1.Introduction to quantum computing



- **The idea of Richard Feynman in May 1981 about kind of computer harnessing quantum physics for solving untractable problem in classic computer like a simulate quantum systems [1]**

"Nature isn't classical, dammit, and if you want to make a simulation of Nature, you'd better make it quantum mechanical, and by golly it's a wonderful problem because it doesn't look so easy"

- **Shor's Algorithm in 1994 [2]**
- **Grover's Search Algorithm in 1995 [3]**
- **VQE algorithm was developed by Peruzzo and McClean et al. in 2014 [4]**

Reference:

- [1] John Preskill. Quantum computing 40 years later. arXiv preprint arXiv:2106.10522,2021.
- [2] Peter W Shor. Algorithms for quantum computation: discrete logarithms and factoring. In Proceedings 35th annual symposium on foundations of computer science, pages 124–134. Ieee, 1994.
- [3] Lov K Grover. Quantum mechanics helps in searching for a needle in a haystack.Physical review letters, 79(2):325, 1997.
- [4] Alberto Peruzzo, Jarrod McClean, Peter Shadbolt, Man-Hong Yung, Xiao-Qi Zhou, Peter J Love, Al'an Aspuru-Guzik, and Jeremy L O'brien. A variational eigenvalue solver on a photonic quantum processor. Nature communications, 5(1):4213, 2014.

1. Introduction to quantum computing



1-qubit state:

$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle$$

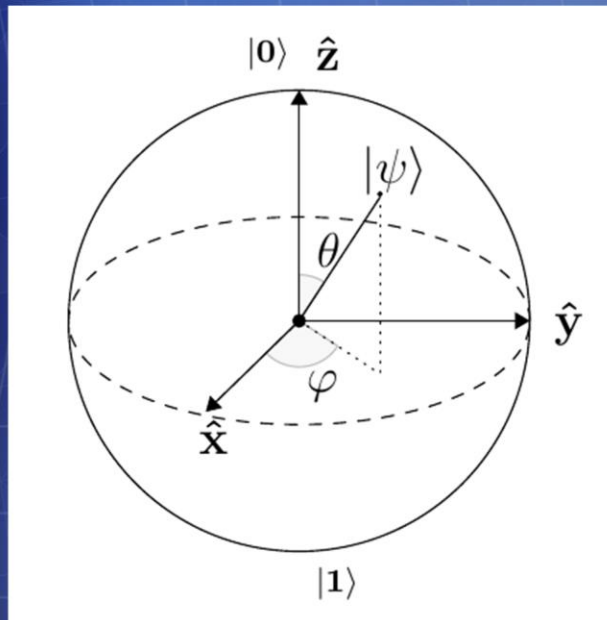
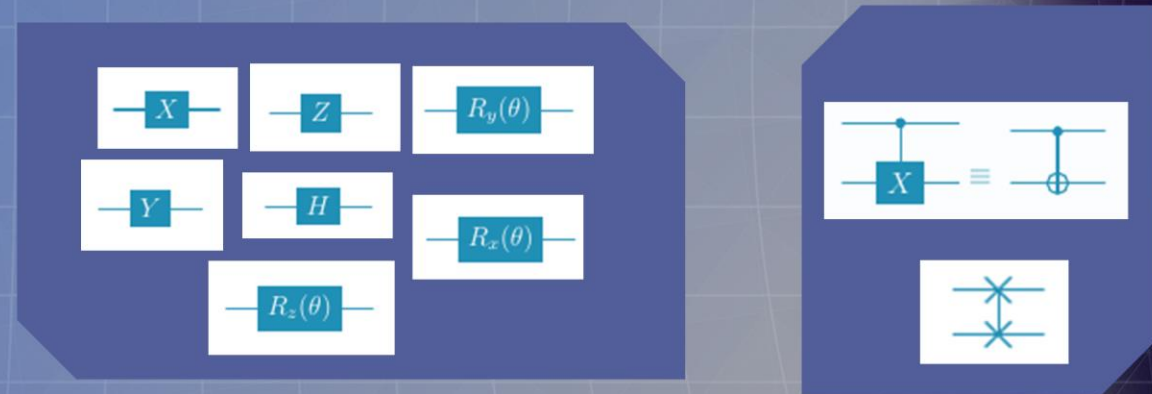


Figure 1.1: Bloch sphere

Quantum gate:

- Single qubit gates: X, Y, Z, H, Rx, Ry, Rz,...
- Two-qubit gates: SWAP, CNOT,...





2. Transverse Ising model

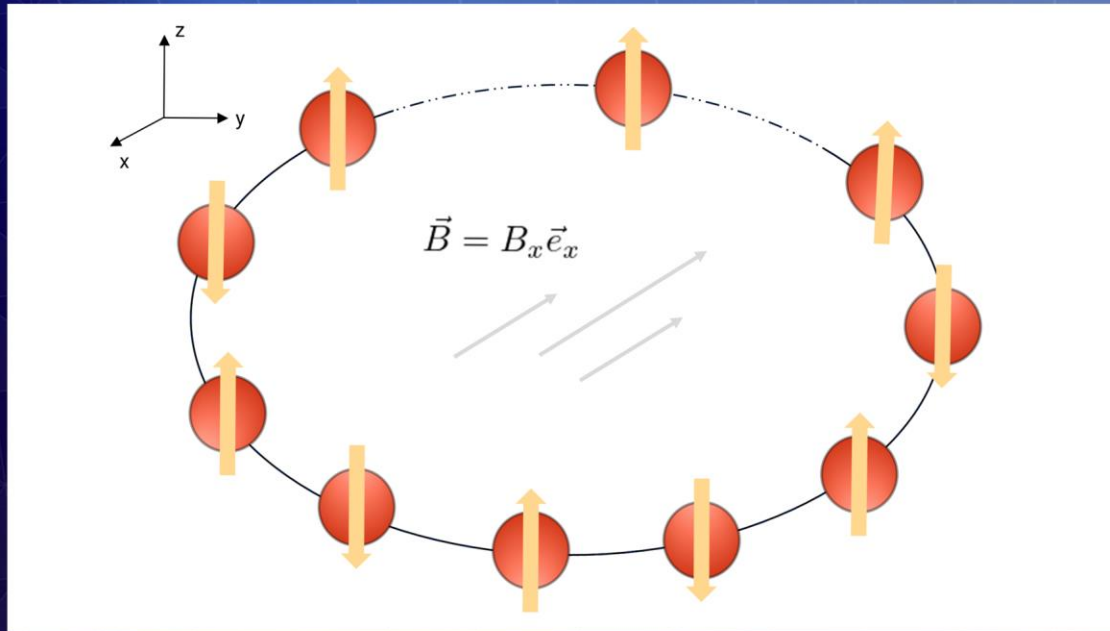


Figure 2.1: Transverse Ising model

Hamiltonian of the transverse ising model

$$H_{\text{TFI}} = -J \sum_i^N \sigma_i^z \sigma_{i+1}^z - h \sum_i^N \sigma_i^x.$$

3. Variational Quantum Eigensolver

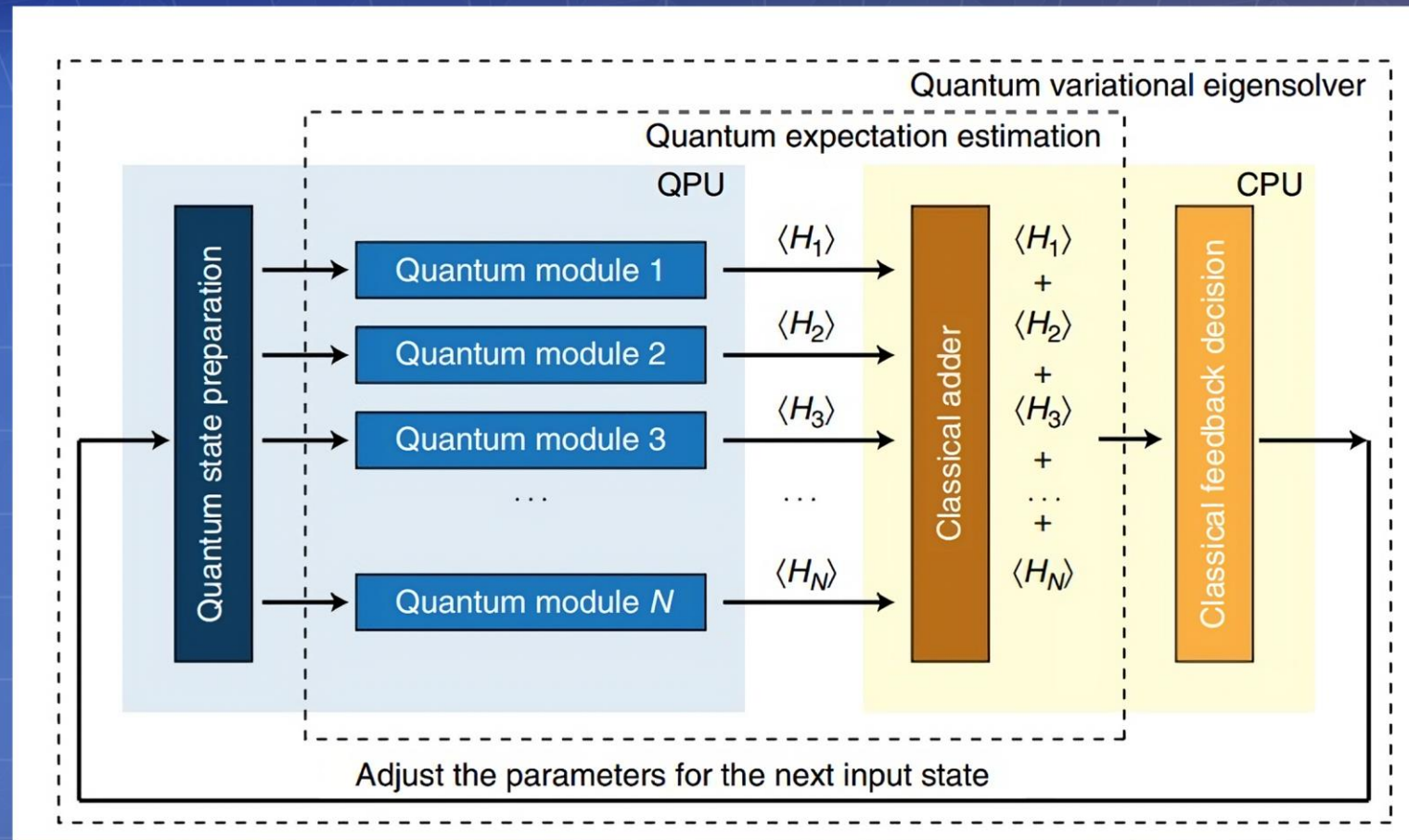
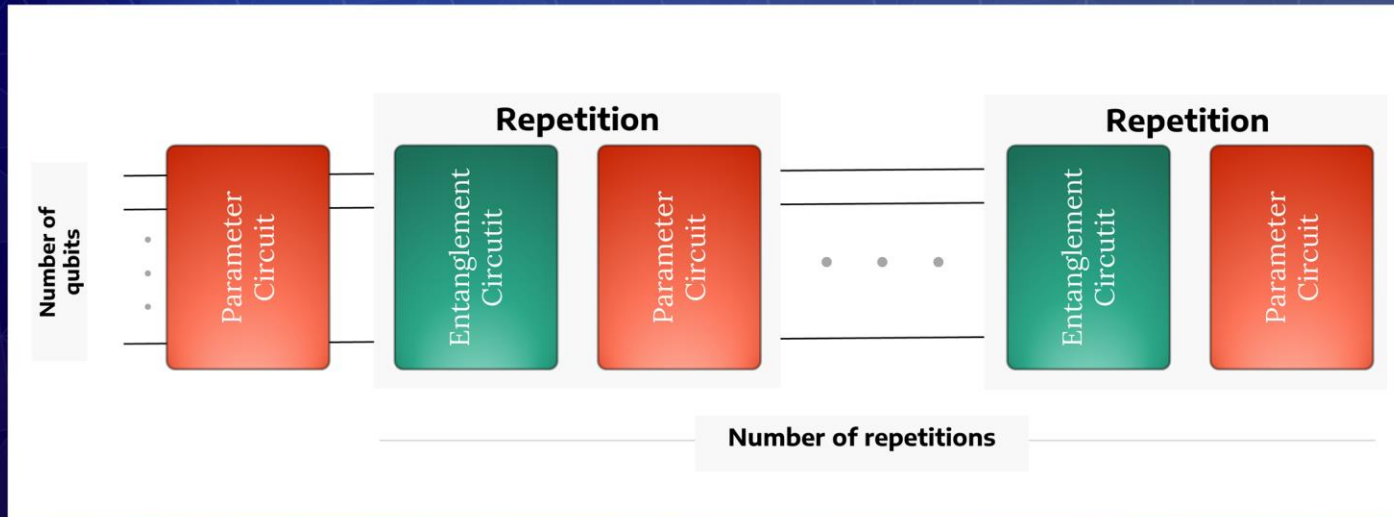


Figure 3.1: Architecture of the quantum-variational eigensolver

Alberto Peruzzo, et al, Nature communications (2014)

3.1 VQE (Ansatz)



For the parameterized circuit, I would use RealAmplitudes ansatz, which is the block of parameter Ry gates, follow with the entanglement circuit.

Figure 3.2: Ansatz structure

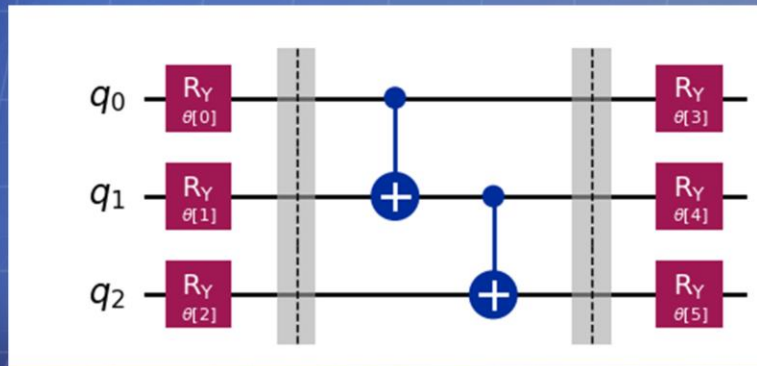


Figure 3.3: RealAmplitudes(3,1) - Linear entanglement

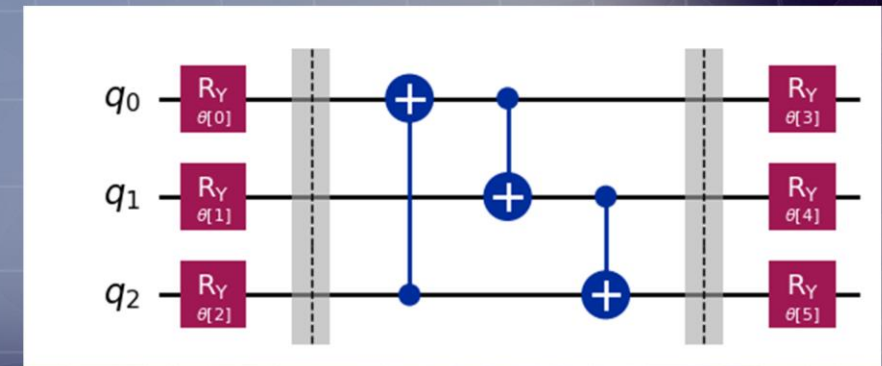


Figure 3.4: RealAmplitudes(3,1) - Full entanglement

3.2 VQE (Measurement)

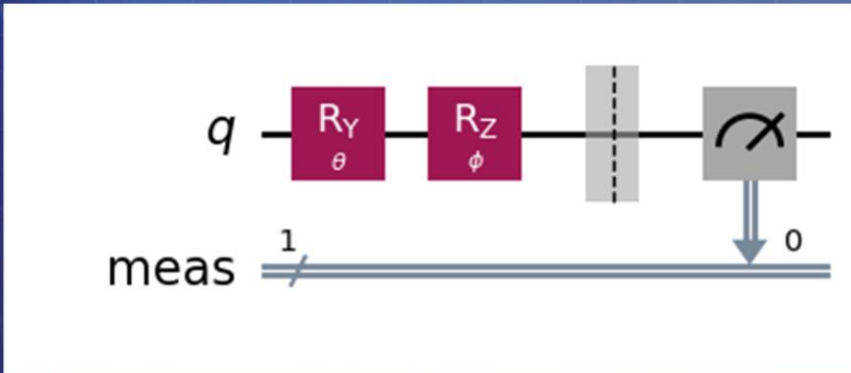


Figure 3.5: Measurement Pauli-Z operator

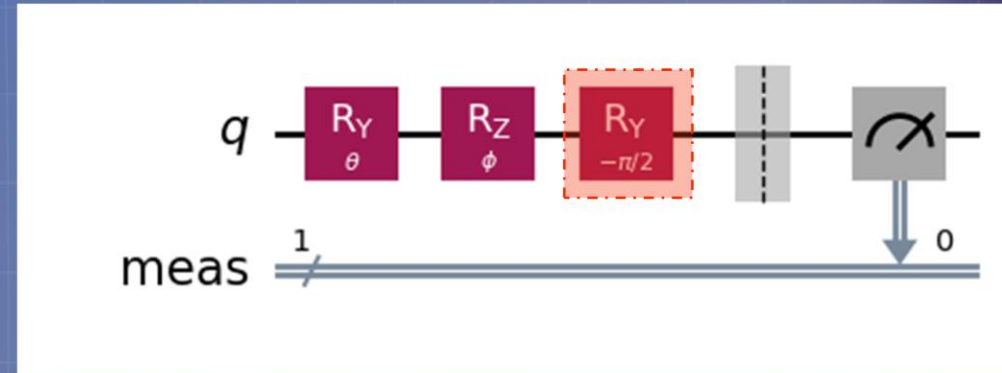


Figure 3.6: Measurement Pauli-X operator

$$\langle \sigma_z \rangle = \langle \psi | (\sigma_z) | \psi \rangle = |a|^2 - |b|^2$$

$$\sigma_x = R_y^\dagger \left(-\frac{\pi}{2} \right) Z R_y \left(-\frac{\pi}{2} \right)$$

3.3 VQE (Measurement)



$$\begin{aligned}\langle \sigma_z^m \sigma_z^n \rangle &= \sum_{q_0, \dots, q_{n-1}=0}^1 \sum_{q'_0, \dots, q'_{n-1}=0}^1 a_{q'_0 \dots q'_{n-1}}^* a_{q_0 \dots q_{n-1}} \langle q'_0 \dots q'_{n-1} | \sigma_z^m \sigma_z^n | q_0 \dots q_{n-1} \rangle, \\&= \sum_{q_0, \dots, q_{n-1}=0}^1 \sum_{q'_0, \dots, q'_{n-1}=0}^1 a_{q'_0 \dots q'_{n-1}}^* a_{q_0 \dots q_{n-1}} \langle q'_0 \dots q'_{n-1} | (-1)^{q_m + q_n} | q_0 \dots q_{n-1} \rangle, \\&= \sum_{q_0, \dots, q_{n-1}=0}^1 \sum_{q'_0, \dots, q'_{n-1}=0}^1 a_{q'_0 \dots q'_{n-1}}^* a_{q_0 \dots q_{n-1}} (-1)^{q_m + q_n} \delta_{q'_0}^{q_0} \dots \delta_{q'_{n-1}}^{q_{n-1}}, \\&= \sum_{q_0, \dots, q_{n-1}=0}^1 |a_{q_0 \dots q_{n-1}}|^2 (-1)^{q_m + q_n}\end{aligned}$$

3.4. Optimization



Gradient descent

$$\theta_{i+1} = \theta_i - \eta \cdot \nabla f(\theta_i)$$



Parameter-shift rule [1]

$$g_i(\theta) = \frac{f(\theta + se_i) - f(\theta - se_i)}{2 \sin(s)}$$

SPSA, QN-SPSA, Nelder-mead

Natural gradient descent

$$\theta_{i+1} = \theta_i - \eta G^{-1} \nabla f(\theta)$$



Quantum natural gradient [2]

$$\begin{aligned} G_{ij}^{(l)} &= \langle \partial_i \psi_\theta | \partial_j \psi_\theta \rangle - (-i \partial_i \langle \psi_\theta | \psi_\theta \rangle) (i \langle \psi_\theta | \partial_j \psi_\theta \rangle) \\ &= \langle \psi_l | K_i K_j | \psi_l \rangle - \langle \psi_l | K_i | \psi_l \rangle \langle \psi_l | K_j | \psi_l \rangle \end{aligned}$$

Reference:

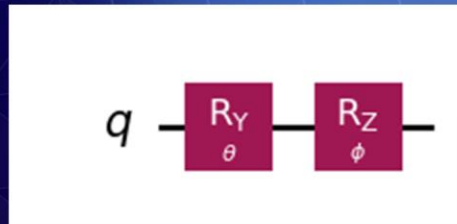
[1] Andrea Mari, et al., Physical Review A, 103(1):012405, 2021.

[2] James Stokes, et al., Quantum natural gradient. Quantum, 4:269, 2020.

4. Implementation and results

n = 1

Ansatz



Parameter state

$$Rz(\phi)Ry(\theta)|0\rangle = \begin{pmatrix} e^{-i\phi/2} \cos(\theta/2) \\ e^{i\phi/2} \sin(\theta/2) \end{pmatrix}$$

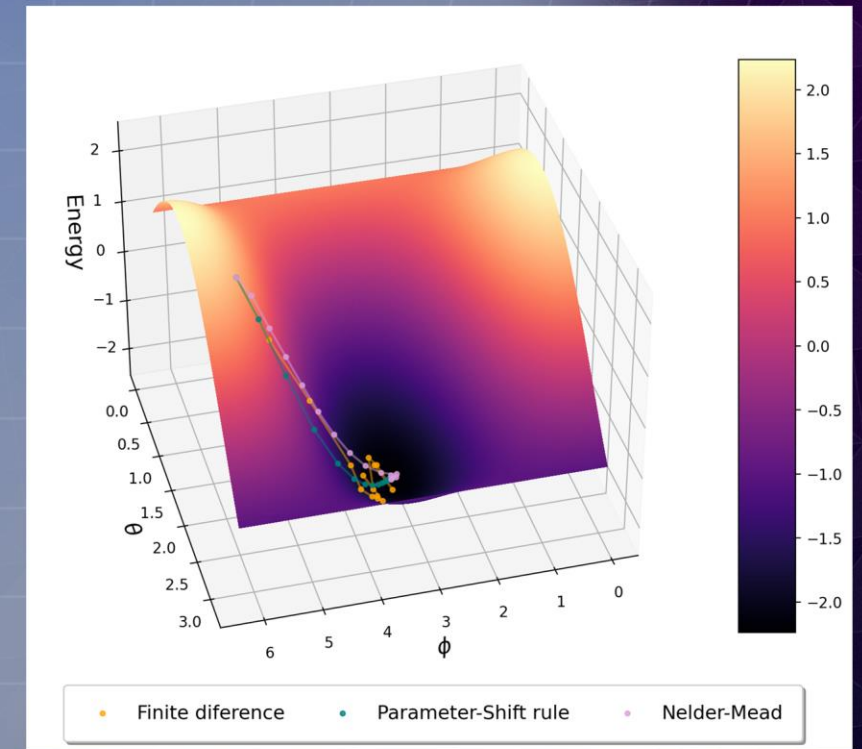
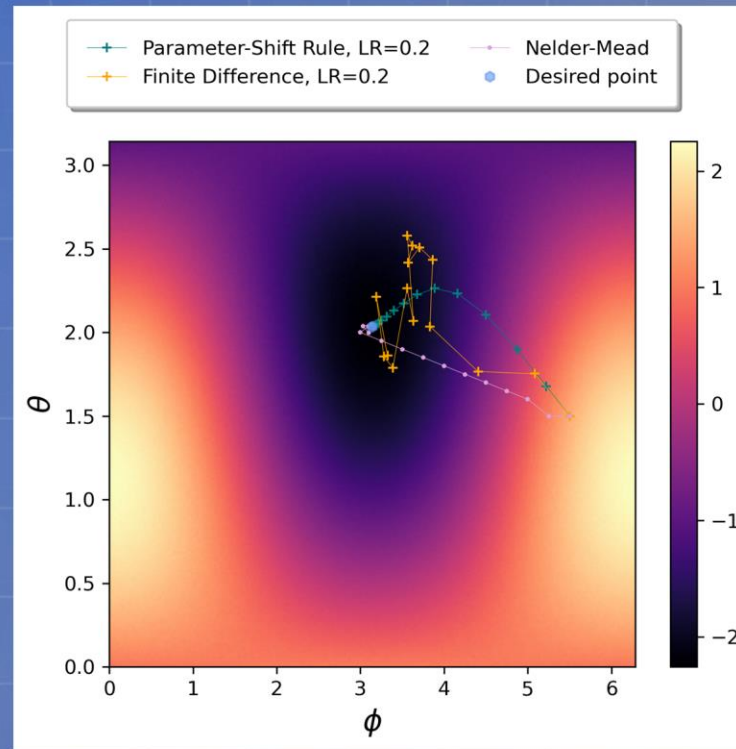


Figure 4.1: Optimization for the loss function having $H = 2X + Z$

4. Implementation and results

$n = 3$

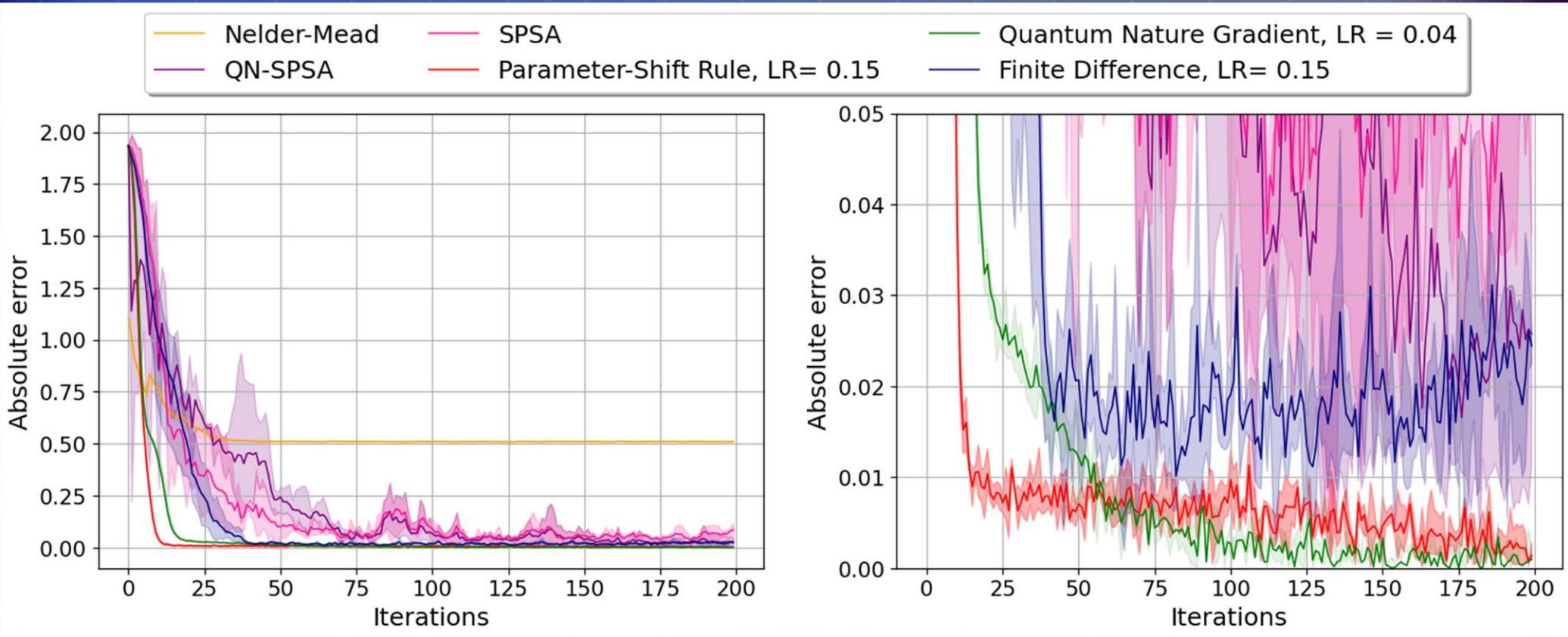


Figure 4.2: Optimization for 3-spin Transverse Ising model

Shots: 20000 shots, iterations: 200 iterations, simulator: qasm_simulator, ansatz: RealAmplitudes(3,2) (Linear entanglement)

4. Implementation and results

$n = 5$

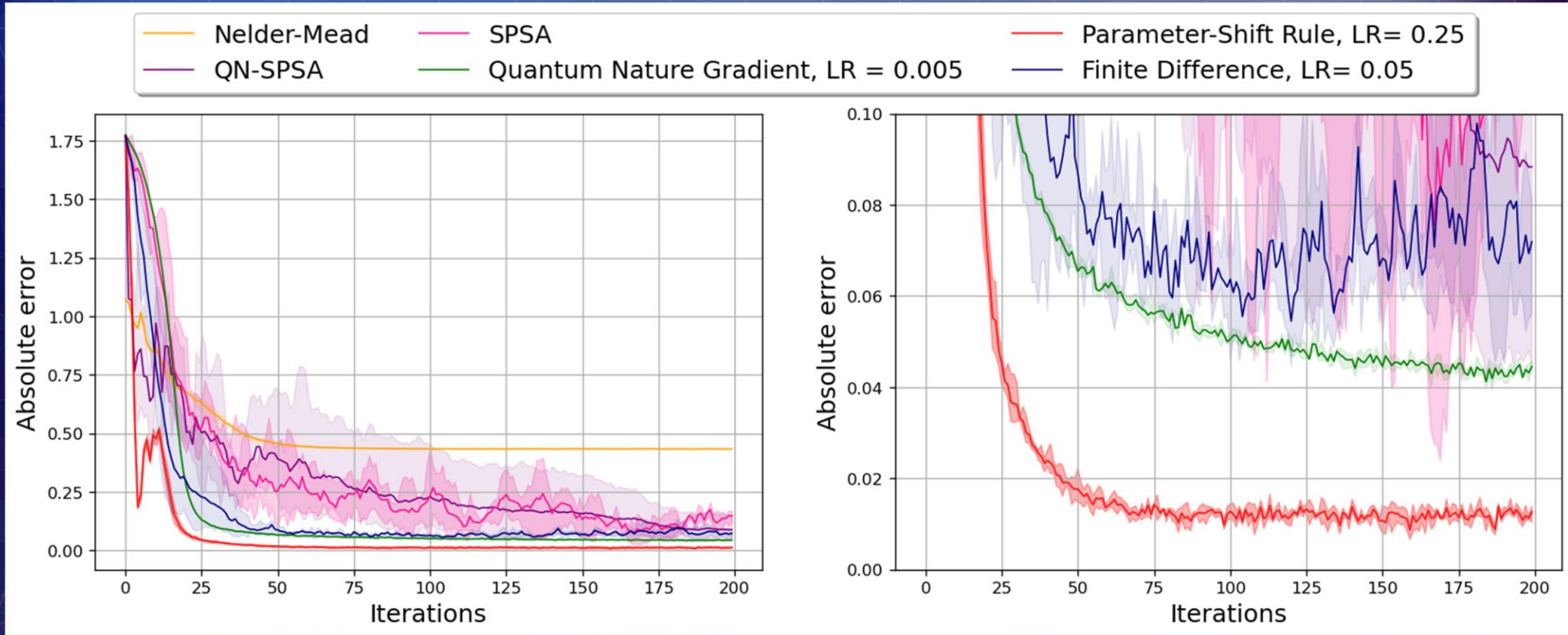


Figure 4.3: Optimization for 5-spin Transverse Ising model

Shots: 20000 shots, iterations: 200 iterations, simulator: qasm_simulator ,
ansatz: RealAmplitudes (5,2) (Linear entanglement)

4. Implementation and results

n = 8

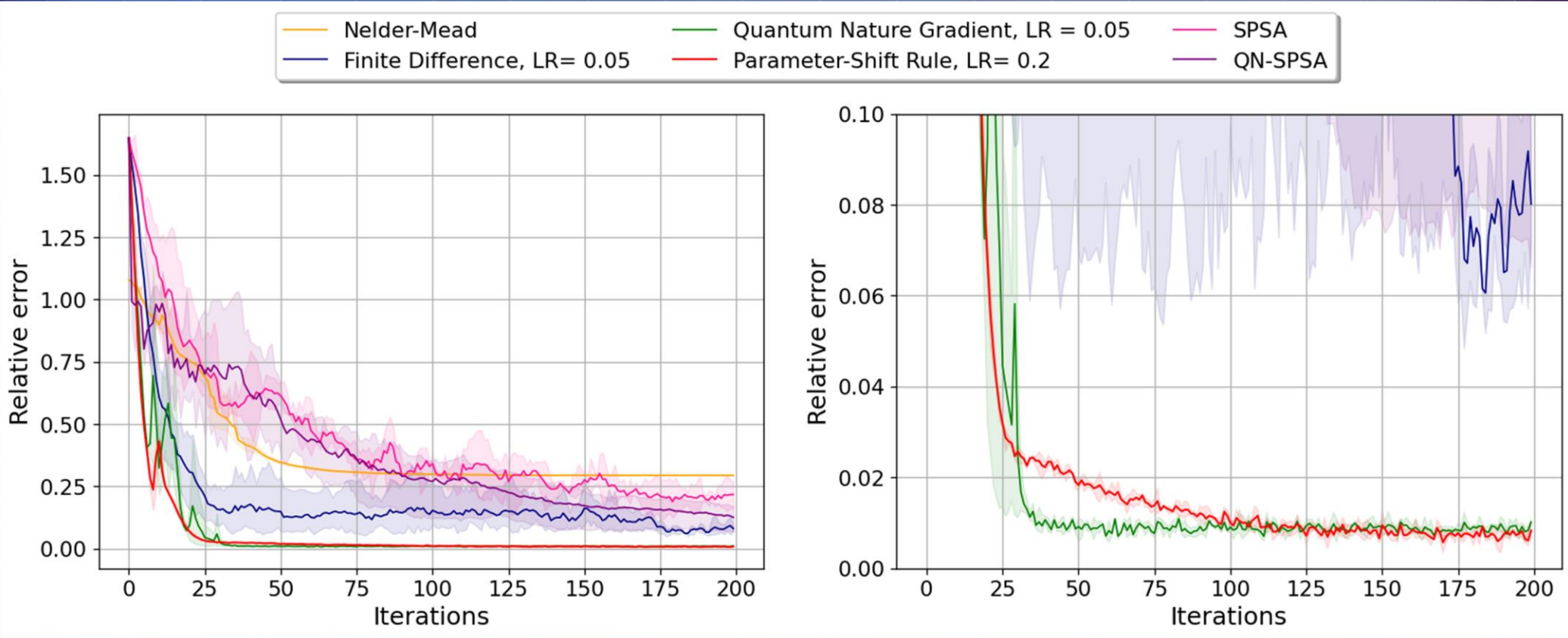


Figure 4.3: Optimization for 8-spin Transverse Ising model

Shots: 20000 shots, iterations: 200 iterations, simulator: qasm_simulator, ansatz: RealAmplitudes (8,2) (Linear entanglement)

4. Implementation and results

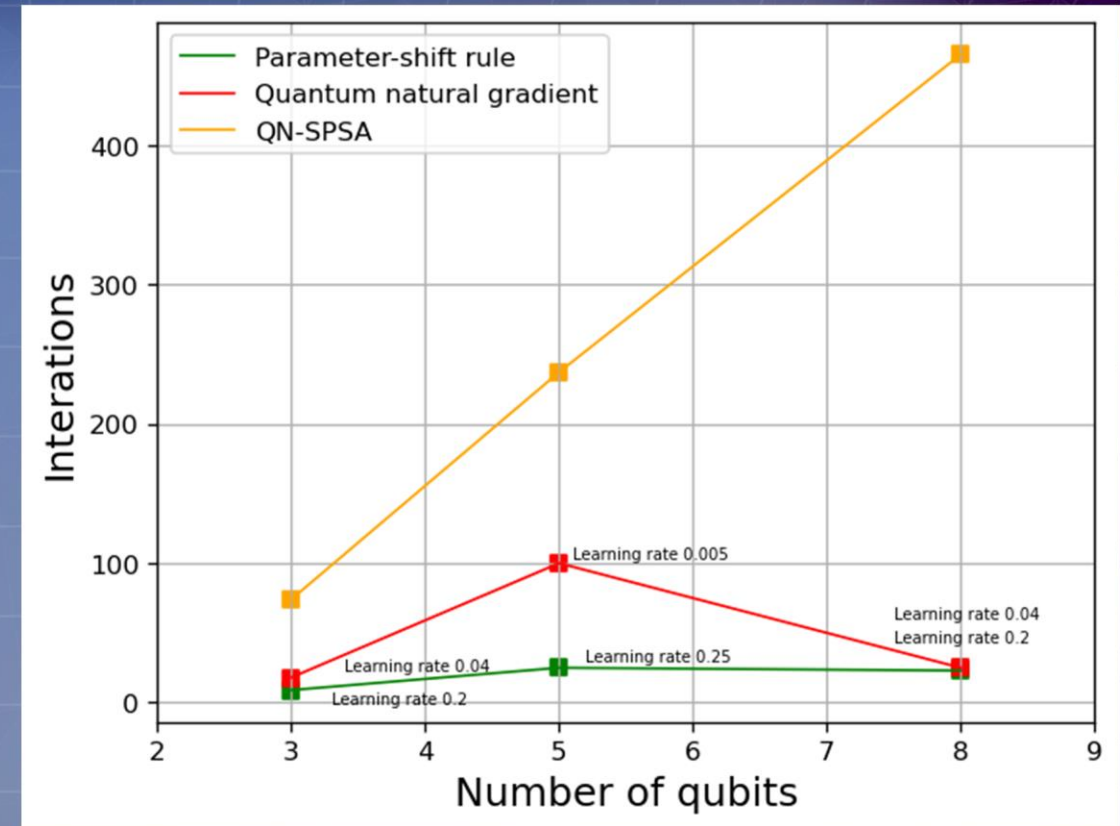
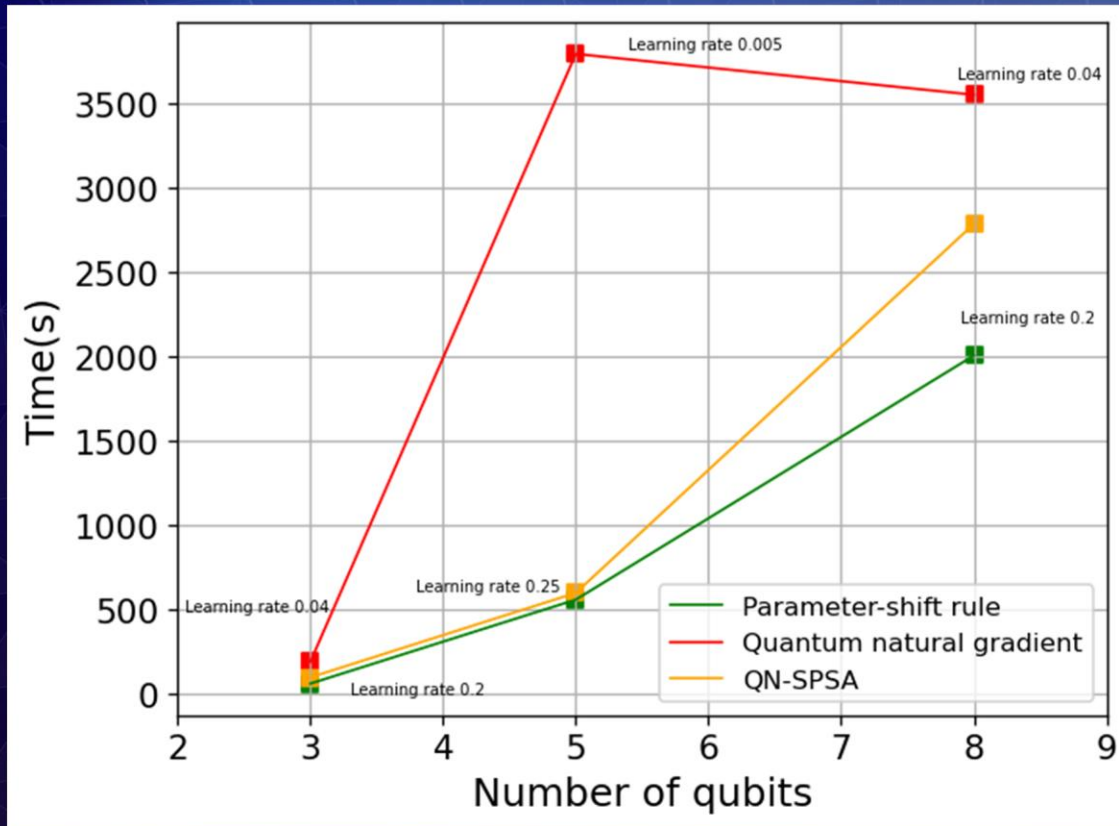
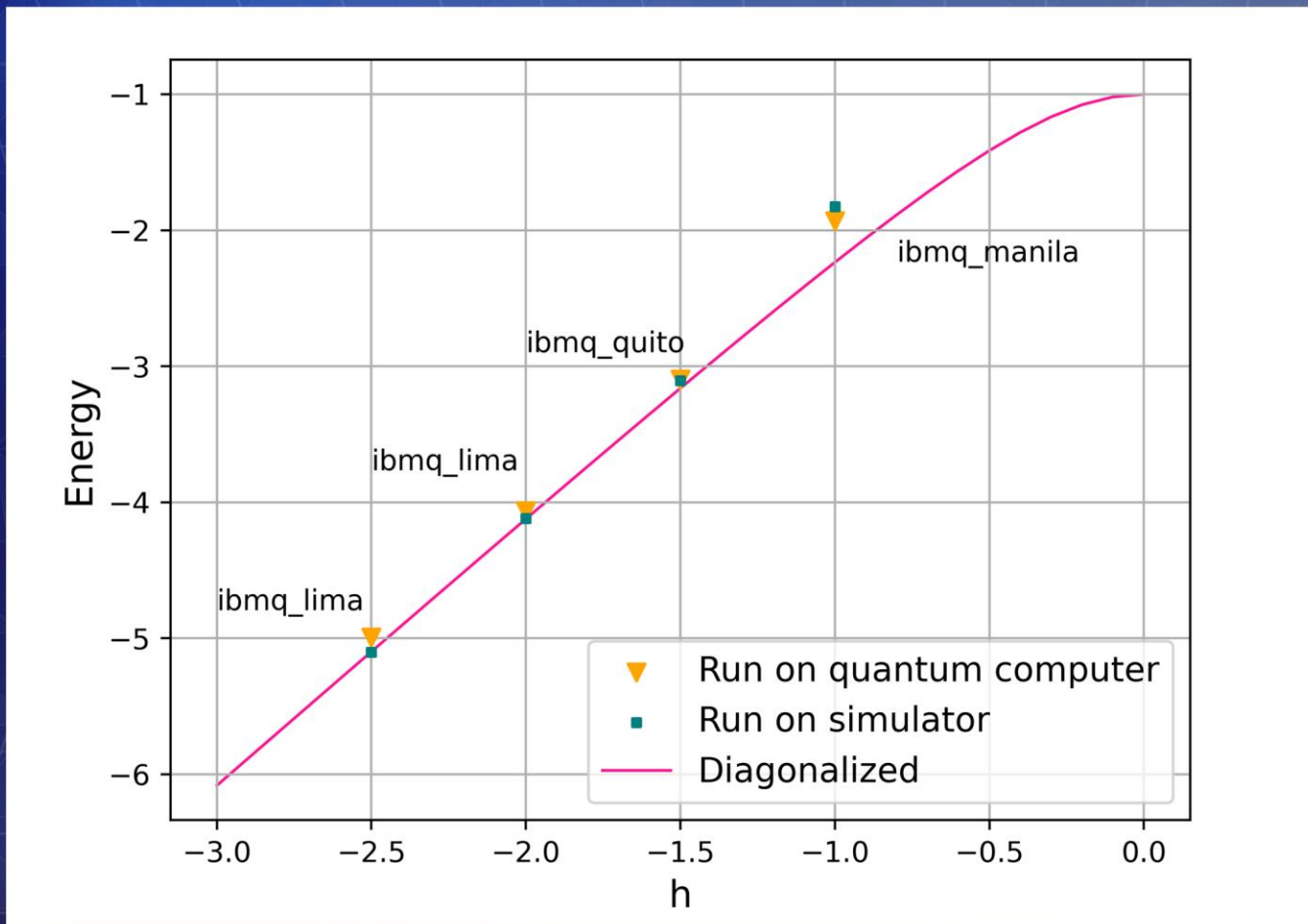


Figure 4.4: Compared number of iterations and runtime of Parameter shift rule, Quantum natural gradient and QN-SPSA

Condition of convergence: 5%



4. Implementation and results



40000 shots
Parameter-shift rule
Learning rate 0.2
20 iteration
RealAmplitudes(2,1)
Transverse Ising model
with 2 spin

**Figure 4.5: Optimization on real quantum computer
ibmq_lima, ibmq_quito, ibmq_manila**



5. Quantum Chemistry with VQE

Classical approach

- **Hatree-Fock method**
- **Post HF method: Configuration Interaction (CI), Couple Cluster, Perturbation Method (MP2),...**

Quantum approach

- **Second quantization**
- **Mapping: Jordan-Wigner, Bravyi-Kitaev**
- **Implement on quantum computer**

Problem with VQE

- **The barren plateau problem [1]**
- **Error mitigation [1]**
- **Ansatz selection & optimization strategies [1]**
- **Measurement [2]**

Reference:

[1] Jules Tilly, et al., Physics Reports, 2022.08.003

[2] Gokhale, et al., doi: {10.1109/QCE49297.2020.00054}



Thank You